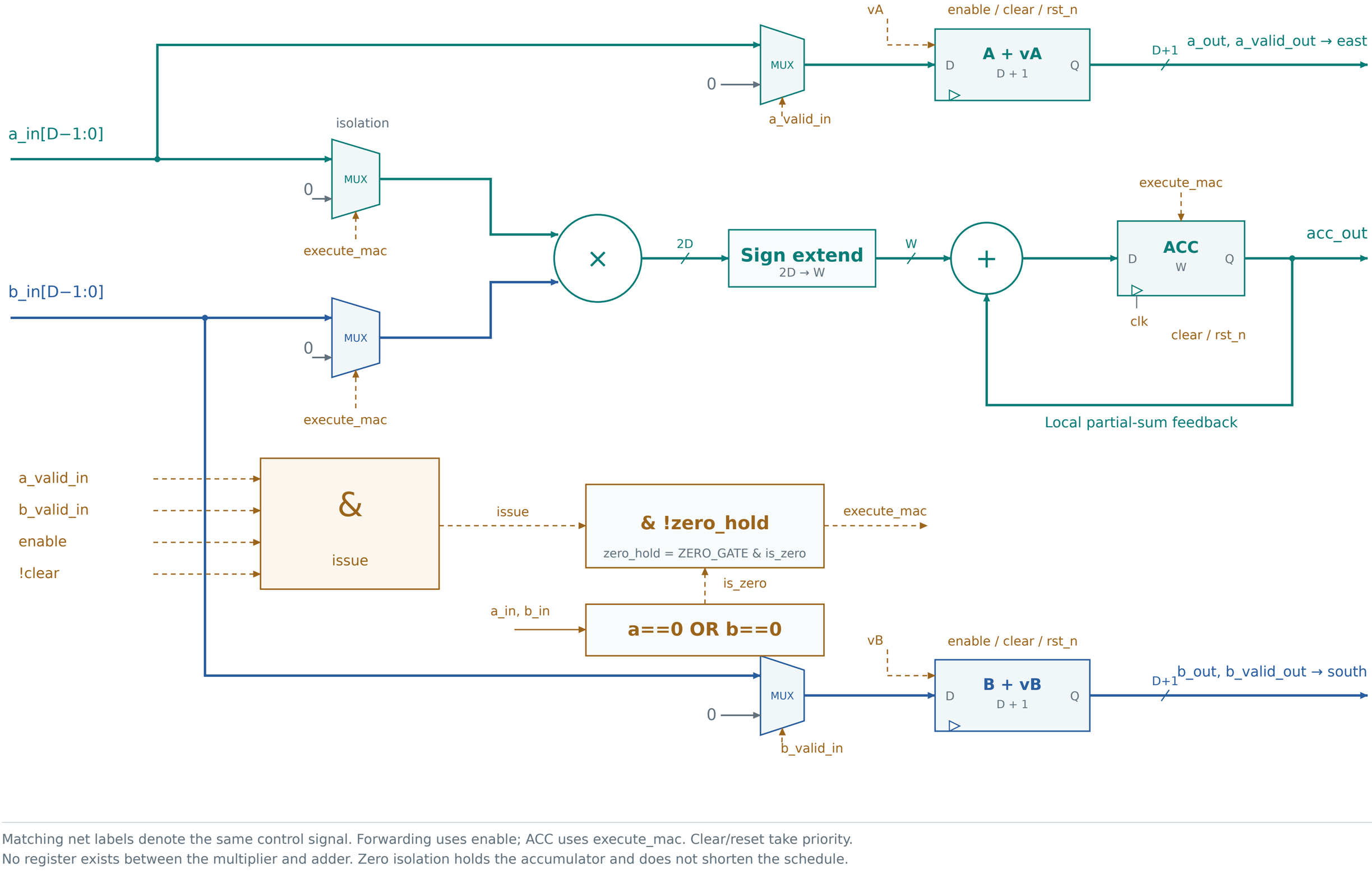
Processing element — register-transfer datapath

systolic_pe + signed_mac · D-bit signed operands · W-bit wrapping accumulator · one registered operand hop per enabled edge



Matching net labels denote the same control signal. Forwarding uses enable; ACC uses execute_mac. Clear/reset take priority.
No register exists between the multiplier and adder. Zero isolation holds the accumulator and does not shorten the schedule.